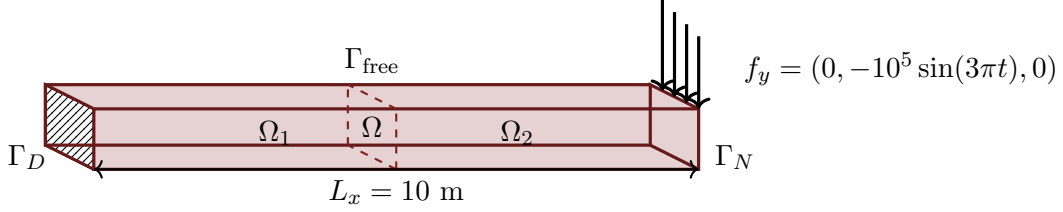


Weak forms and problem setup — 3D beam under time-dependent transverse traction

1 Governing equations

1.1 Geometry, regions, and boundary conditions



Let the beam occupy

$$\Omega := (0, L_x) \times (0, L_y) \times (0, L_z) \subset \mathbb{R}^3, \quad (L_x, L_y, L_z) = (10, 1, 1).$$

The boundary is decomposed into

$$\begin{aligned} \Gamma_D &:= \{(x, y, z) \in \partial\Omega : x = 0\}, \\ \Gamma_N &:= \{(x, y, z) \in \partial\Omega : x = L_x\}, \\ \Gamma_{\text{free}} &:= \partial\Omega \setminus (\Gamma_D \cup \Gamma_N). \end{aligned}$$

The two material subdomains are

$$\begin{aligned} \Omega_1 &:= \{(x, y, z) \in \Omega : x \leq L_x/2\}, \\ \Omega_2 &:= \{(x, y, z) \in \Omega : x > L_x/2\}. \end{aligned}$$

In the implementation, these are stored as `region_1` and `region_2`.

1.2 Material parameterization and damping

The sampled parameters are Young's modulus E and Poisson ratio ν ,

$$E \in [3 \times 10^6, 10 \times 10^6], \quad \nu \in [0.20, 0.40].$$

The baseline Lamé parameters are

$$\lambda(E, \nu) = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}, \quad \mu(E, \nu) = \frac{E}{2(1 + \nu)}.$$

The regionwise coefficients used in the code are

$$(\lambda_1, \mu_1) = 50(\lambda, \mu), \quad (\lambda_2, \mu_2) = (\lambda, \mu).$$

The density is

$$\rho = 7850.$$

Rayleigh damping is introduced after spatial discretization as

$$\mathbf{C} = c_v \mathbf{M} + c_m \mathbf{K}, \quad c_v = 10^{-3}, \quad c_m = 10^{-1}.$$

1.3 Kinematics and constitutive law

Let $\mathbf{u} : \Omega \times [0, T] \rightarrow \mathbb{R}^3$ denote the displacement field. The infinitesimal strain tensor is

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top).$$

In each region Ω_r , $r = 1, 2$, the linear elastic Cauchy stress is

$$\boldsymbol{\sigma}_r(\mathbf{u}) = 2\mu_r \boldsymbol{\varepsilon}(\mathbf{u}) + \lambda_r \operatorname{tr}(\boldsymbol{\varepsilon}(\mathbf{u})) \mathbf{I}.$$

1.4 Strong form

Let $T = 120$. The code uses a discrete time grid

$$t_i \in [0, 120], \quad i = 0, \dots, 1999.$$

Ignoring Rayleigh damping at the continuous level, the transient elastodynamic problem is: find $\mathbf{u}$ such that

$$\rho \ddot{\mathbf{u}} - \nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{0} \quad \text{in } \Omega \times (0, T], \quad (1)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_D \times (0, T], \quad (2)$$

$$\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = \mathbf{t}(t) \quad \text{on } \Gamma_N \times (0, T], \quad (3)$$

$$\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_{\text{free}} \times (0, T], \quad (4)$$

with

$$\mathbf{t}(t) = (0, -10^5 \sin(3\pi t), 0).$$

The initial displacement and velocity are taken as zero on the free degrees of freedom.

2 Weak form and semi-discrete system

2.1 Variational statement

Define the admissible space

$$V := \{\mathbf{v} \in [H^1(\Omega)]^3 \mid \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D\}.$$

The weak problem reads: find $\mathbf{u}(t) \in V$ such that, for all $\mathbf{v} \in V$,

$$m(\ddot{\mathbf{u}}, \mathbf{v}) + a(\mathbf{u}, \mathbf{v}; E, \nu) = \ell(\mathbf{v}; t),$$

where

$$m(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \rho \mathbf{u} \cdot \mathbf{v} \, d\Omega, \quad (5)$$

$$a(\mathbf{u}, \mathbf{v}; E, \nu) = \sum_{r=1}^2 \int_{\Omega_r} [2\mu_r \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) + \lambda_r \operatorname{tr}(\boldsymbol{\varepsilon}(\mathbf{u})) \operatorname{tr}(\boldsymbol{\varepsilon}(\mathbf{v}))] \, d\Omega, \quad (6)$$

and

$$\ell(\mathbf{v}; t) = \int_{\Gamma_N} -10^5 \sin(3\pi t) v_y \, d\Gamma. \quad (7)$$

2.2 Finite element semi-discrete model

Using vector-valued trilinear hexahedral elements, the semi-discrete system solved in the code is

$$\mathbf{M}_{ff}\ddot{\mathbf{u}}_{ff} + \mathbf{C}_{ff}\dot{\mathbf{u}}_{ff} + \mathbf{K}_{ff}(E, \nu)\mathbf{u}_{ff} = \mathbf{f}_{ff}(t), \quad (8)$$

where the subscript ff denotes the free degrees of freedom after elimination of the clamped left-face degrees of freedom.

The damping matrix is

$$\mathbf{C}_{ff} = c_v \mathbf{M}_{ff} + c_m \mathbf{K}_{ff}.$$

Time integration is performed with a Newmark scheme.

2.3 Affine decomposition

The stiffness operator admits a regionwise affine decomposition:

$$\mathbf{K}(E, \nu) = \sum_{r=1}^2 [\lambda_r(E, \nu) \mathbf{K}_{\lambda,r} + \mu_r(E, \nu) \mathbf{K}_{\mu,r}]. \quad (9)$$

The four parameter-independent stiffness blocks correspond to the following contributions:

Block	Coefficient	Bilinear contribution
$\mathbf{K}_{\lambda,r}$	$\lambda_r(E, \nu)$	$\int_{\Omega_r} \text{tr}(\boldsymbol{\varepsilon}(\mathbf{u})) \text{tr}(\boldsymbol{\varepsilon}(\mathbf{v})) \, d\Omega$
$\mathbf{K}_{\mu,r}$	$\mu_r(E, \nu)$	$\int_{\Omega_r} 2 \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega$

Here $r = 1, 2$, with region 1 scaled by a factor of 50 and region 2 unscaled. This is the affine structure used in the full-order assembly.